CYBERSRS

Secure DNS and Web Access Filtering System

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure DNS and web-access filtering system for a university that protects students, faculty, staff, and administrative users from malicious or unauthorized internet destinations. The system should inspect DNS requests, block access to known malicious domains, apply different browsing policies based on user roles, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators should be able to create and update filtering policies, allow approved exceptions, investigate repeated attempts to access blocked resources, and temporarily restrict users when suspicious behavior is detected. The platform should maintain audit logs of policy changes and filtering decisions, integrate with centralized authentication, and support deployment across academic buildings, hostels, laboratories, and administrative offices. The solution should be secure, scalable, reliable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure DNS and web-access filtering system for a university that protects students, faculty, staff, and administrative users from malicious or unauthorized internet destinations.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should inspect DNS requests, block access to known malicious domains, apply different browsing policies based on user roles, and provide administrators with a dashboard for reviewing blocked requests and security events.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Administrators should be able to create and update filtering policies, allow approved exceptions, investigate repeated attempts to access blocked resources, and temporarily restrict users when suspicious behavior is detected.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The platform should maintain audit logs of policy changes and filtering decisions, integrate with centralized authentication, and support deployment across academic buildings, hostels, laboratories, and administrative offices.

Status: confirmed

Evidence ids: E-004

Stakeholder source statement E-005

Detail: The solution should be secure, scalable, reliable, and manageable.

Status: confirmed

Evidence ids: E-005

Open issues: how user roles are defined and mapped to browsing policies; what constitutes suspicious behavior and how it is detected; how audit logs are stored and accessed by authorized personnel; whether the system will support real-time threat intelligence feeds

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - implement a secure DNS and web-access filtering mechanism to protect all university users from malicious or unauthorized internet destinations.

Statement: The system shall implement a secure DNS and web-access filtering mechanism to protect all university users from malicious or unauthorized internet destinations.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A student attempts to access a known malicious domain

When: The DNS resolver receives the request

Then: The system blocks the request.

Method: test

Observable evidence: Test capture of the resulting state

Unresolved decisions: Definition of 'malicious domain'

FR-002 - inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events.

Statement: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A faculty member attempts to access a restricted educational resource

When: The system evaluates the request against role-based policy rules

Then: Access is granted according to the faculty policy

Method: test

Observable evidence: Test capture of policy evaluation and access decision

Given: An administrator initiates a dashboard query for blocked requests

When: The system retrieves security event logs from its internal store

Then: The dashboard returns a list of recent blocked domains and associated user actions

Method: inspection

Observable evidence: Test capture of dashboard response to query

Unresolved decisions: Threshold for repeated access attempts; Definition of 'known malicious domain'

FR-003 - Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Statement: Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Rationale: Implements stakeholder source E-003.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: An administrator creates a new filtering policy for staff

When: The system receives the policy configuration request

Then: The policy is stored and applied to staff user sessions

Method: demonstration

Observable evidence: Test capture of policy creation and application confirmation

Given: The system detects repeated access attempts

When: The system detects repeated access attempts

Then: The system triggers an investigation workflow

Method: analysis

Observable evidence: Test capture of the observed result

Given: An administrator approves an exception for a specific domain

When: The system receives the exception request

Then: Access to that domain is permitted for the specified user or group

Method: test

Observable evidence: Test capture of approved exception and access grant

Unresolved decisions: Definition of 'suspicious behavior'; Duration of temporary restriction; Workflow for investigation review

3. Non-functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

NFR-001 - The solution shall be secure, scalable, reliable, and manageable.

Statement: The solution shall be secure, scalable, reliable, and manageable.

Rationale: Implements stakeholder source E-005.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-005

Acceptance criteria

Given: A system under operational load with concurrent user sessions

When: The system receives a request to scale resource allocation dynamically

Then: The system maintains service availability without degradation in performance or data integrity

Method: test

Observable evidence: Test capture of system response during dynamic scaling operation

Given: An administrator initiates a configuration change to the security policy

When: The system applies the updated policy to active user sessions

Then: The updated security policy is applied to active user sessions

Method: inspection

Observable evidence: Test capture of the resulting state

Given: A component failure is detected in the core service layer

When: The system initiates failover to redundant components

Then: The system restores functionality within defined recovery time objectives without data loss

Method: demonstration

Observable evidence: Test capture of failover sequence and service continuity

Given: An operator attempts to modify configuration beyond authorized scope

When: The system enforces access control policies during management operations

Then: The system denies the action.

Method: analysis

Observable evidence: Test capture of the observed result

Unresolved decisions: Definition of 'manageable' in terms of administrative interface usability thresholds; Thresholds for scalability under peak load conditions; Recovery time objectives for component failure scenarios

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003

5. Data requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

DR-001 - The platform shall maintain audit logs of policy changes and filtering decisions, integrate with centralized authentication, and support deployment across academic buildings, hostels, laboratories, and administrative offices.

Statement: The platform shall maintain audit logs of policy changes and filtering decisions, integrate with centralized authentication, and support deployment across academic buildings, hostels, laboratories, and administrative offices.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: data

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A system administrator modifies a filtering policy in the university network

When: The policy change is applied to the network configuration

Then: An audit log entry is generated recording the timestamp, user identity, action type, and affected resource scope

Method: test

Observable evidence: External test capture of the audit log response after policy modification

Given: A user attempts to access a domain blocked by role-based filtering policies

When: The system evaluates the request against active policies and authentication status

Then: The system either blocks the access or allows it based on defined rules, with a decision recorded in audit logs

Method: inspection

Observable evidence: External test capture of filtering decision outcome and corresponding log entry

Given: Centralized authentication service returns a user identity for an academic building network access request

When: The system receives the identity via integration with centralized authentication

Then: The system validates the identity and applies appropriate role-based policies without requiring manual intervention

Method: demonstration

Observable evidence: External test capture of successful authentication flow and policy application

Given: The platform is deployed in an academic building, hostel, laboratory, or administrative office environment

When: Network traffic is observed across these physical locations

Then: The system maintains consistent filtering and policy enforcement without degradation in performance or availability

Method: analysis

Observable evidence: External test capture of network behavior before and after deployment in each location

Unresolved decisions: Whether audit logs shall include metadata about the source IP address of policy change initiators; The exact scope of role-based policies across different building types; How policy synchronization is handled during cross-location deployments

Related requirement ids: FR-002

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: DR-001; FR-001

7. Compliance and policy applicability matrix

Open decisions: how user roles are defined and mapped to browsing policies; what constitutes suspicious behavior and how it is detected; how audit logs are stored and accessed by authorized personnel; whether the system will support real-time threat intelligence feeds

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: administrators

Detail: Proposed logical endpoint for the source-stated actor: administrators.

Status: proposed

Evidence ids: E-002; E-003

Requirement ids: FR-002; FR-003

Component name: Actor: administrators

Responsibility: Represent the source-stated actor: administrators.

Policy Engine

Detail: Proposed logical owner for FR-002, FR-003, NFR-001.

Status: proposed

Evidence ids: E-002; E-003; E-005

Requirement ids: FR-002; FR-003; NFR-001

Component name: Policy Engine

Responsibility: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected. The solution shall be secure, scalable, reliable, and manageable.

DNS and Web Access Filter

Detail: Proposed logical owner for FR-001, DR-001, NFR-001.

Status: proposed

Evidence ids: E-001; E-004; E-005

Requirement ids: FR-001; DR-001; NFR-001

Component name: DNS and Web Access Filter

Responsibility: The system shall implement a secure DNS and web-access filtering mechanism to protect all university users from malicious or unauthorized internet destinations. The platform shall maintain audit logs of policy changes and filtering decisions, integrate with centralized authentication, and support deployment across academic buildings, hostels, laboratories, and administrative offices. The solution shall be secure, scalable, reliable, and manageable.

Administrator Dashboard

Detail: Proposed logical owner for FR-002, FR-003, NFR-001.

Status: proposed

Evidence ids: E-002; E-003; E-005

Requirement ids: FR-002; FR-003; NFR-001

Component name: Administrator Dashboard

Responsibility: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected. The solution shall be secure, scalable, reliable, and manageable.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003.

Status: proposed

Evidence ids: E-002; E-003

Requirement ids: FR-002; FR-003

From component: Actor: administrators

To component: Policy Engine

Data or action: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-005

Requirement ids: NFR-001

From component: Policy Engine

To component: DNS and Web Access Filter

Data or action: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, NFR-001.

Status: proposed

Evidence ids: E-002; E-003; E-005

Requirement ids: FR-002; FR-003; NFR-001

From component: Policy Engine

To component: Administrator Dashboard

Data or action: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected. The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-005

Requirement ids: NFR-001

From component: DNS and Web Access Filter

To component: Policy Engine

Data or action: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-005

Requirement ids: NFR-001

From component: DNS and Web Access Filter

To component: Administrator Dashboard

Data or action: The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, NFR-001.

Status: proposed

Evidence ids: E-002; E-003; E-005

Requirement ids: FR-002; FR-003; NFR-001

From component: Administrator Dashboard

To component: Policy Engine

Data or action: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected. The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-005

Requirement ids: NFR-001

From component: Administrator Dashboard

To component: DNS and Web Access Filter

Data or action: The solution shall be secure, scalable, reliable, and manageable.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003.

Status: proposed

Evidence ids: E-002; E-003

Requirement ids: FR-002; FR-003

Untrusted component: Actor: administrators

Trusted component: Policy Engine

Enforcement: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003, NFR-001.

Status: proposed

Evidence ids: E-002; E-003; E-005

Requirement ids: FR-002; FR-003; NFR-001

Untrusted component: Administrator Dashboard

Trusted component: Policy Engine

Enforcement: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events. Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected. The solution shall be secure, scalable, reliable, and manageable.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-005

Requirement ids: NFR-001

Untrusted component: Administrator Dashboard

Trusted component: DNS and Web Access Filter

Enforcement: The solution shall be secure, scalable, reliable, and manageable.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: Administrators

Goal: Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Story: As Administrators, I need the following behavior: Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Preconditions

1. The system detects repeated access attempts

Steps

1. The system detects repeated access attempts

Success outcome: The system triggers an investigation workflow

Alternate flows

1. Given: An administrator creates a new filtering policy for staff When: The system receives the policy configuration request Then: The policy is stored and applied to staff user sessions Evidence: Test capture of policy creation and application confirmation

Requirement ids: FR-003

Evidence ids: E-003

Unresolved decisions: Definition of 'suspicious behavior'; Duration of temporary restriction; Workflow for investigation review

Id: UC-002

Actor: Administrators

Goal: Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Story: As Administrators, I need the following behavior: Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.

Preconditions

1. An administrator approves an exception for a specific domain

Steps

1. The system receives the exception request

Success outcome: Access to that domain is permitted for the specified user or group

Requirement ids: FR-003

Evidence ids: E-003

Unresolved decisions: Definition of 'suspicious behavior'; Duration of temporary restriction; Workflow for investigation review

Open issues: Definition of 'suspicious behavior'; Duration of temporary restriction; Workflow for investigation review

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A student attempts to access a known malicious domain
2. When: The DNS resolver receives the request

Expected result: Expected: The system blocks the request.

Evidence to record: Evidence: Test capture of the resulting state

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: A faculty member attempts to access a restricted educational resource
2. When: The system evaluates the request against role-based policy rules

Expected result: Expected: Access is granted according to the faculty policy

Evidence to record: Evidence: Test capture of policy evaluation and access decision

Test type: positive

Id: TEST-003

Requirement ids: FR-002

Procedure

1. Given: An administrator initiates a dashboard query for blocked requests
2. When: The system retrieves security event logs from its internal store

Expected result: Expected: The dashboard returns a list of recent blocked domains and associated user actions

Evidence to record: Evidence: Test capture of dashboard response to query

Test type: inspection

Id: TEST-004

Requirement ids: FR-003

Procedure

1. Given: An administrator creates a new filtering policy for staff
2. When: The system receives the policy configuration request

Expected result: Expected: The policy is stored and applied to staff user sessions

Evidence to record: Evidence: Test capture of policy creation and application confirmation

Test type: positive

Id: TEST-005

Requirement ids: FR-003

Procedure

1. Given: The system detects repeated access attempts
2. When: The system detects repeated access attempts

Expected result: Expected: The system triggers an investigation workflow

Evidence to record: Evidence: Test capture of the observed result

Test type: analysis

Id: TEST-006

Requirement ids: FR-003

Procedure

1. Given: An administrator approves an exception for a specific domain
2. When: The system receives the exception request

Expected result: Expected: Access to that domain is permitted for the specified user or group

Evidence to record: Evidence: Test capture of approved exception and access grant

Test type: positive

Id: TEST-007

Requirement ids: NFR-001

Procedure

1. Given: A system under operational load with concurrent user sessions
2. When: The system receives a request to scale resource allocation dynamically

Expected result: Expected: The system maintains service availability without degradation in performance or data integrity

Evidence to record: Evidence: Test capture of system response during dynamic scaling operation

Test type: positive

Id: TEST-008

Requirement ids: NFR-001

Procedure

1. Given: An administrator initiates a configuration change to the security policy
2. When: The system applies the updated policy to active user sessions

Expected result: Expected: The updated security policy is applied to active user sessions

Evidence to record: Evidence: Test capture of the resulting state

Test type: inspection

Id: TEST-009

Requirement ids: NFR-001

Procedure

1. Given: A component failure is detected in the core service layer
2. When: The system initiates failover to redundant components

Expected result: Expected: The system restores functionality within defined recovery time objectives without data loss

Evidence to record: Evidence: Test capture of failover sequence and service continuity

Test type: positive

Id: TEST-010

Requirement ids: NFR-001

Procedure

1. Given: An operator attempts to modify configuration beyond authorized scope
2. When: The system enforces access control policies during management operations

Expected result: Expected: The system denies the action.

Evidence to record: Evidence: Test capture of the observed result

Test type: analysis

Id: TEST-011

Requirement ids: DR-001

Procedure

1. Given: A system administrator modifies a filtering policy in the university network
2. When: The policy change is applied to the network configuration

Expected result: Expected: An audit log entry is generated recording the timestamp, user identity, action type, and affected resource scope

Evidence to record: Evidence: External test capture of the audit log response after policy modification

Test type: positive

Id: TEST-012

Requirement ids: DR-001

Procedure

1. Given: A user attempts to access a domain blocked by role-based filtering policies
2. When: The system evaluates the request against active policies and authentication status

Expected result: Expected: The system either blocks the access or allows it based on defined rules, with a decision recorded in audit logs

Evidence to record: Evidence: External test capture of filtering decision outcome and corresponding log entry

Test type: inspection

Id: TEST-013

Requirement ids: DR-001

Procedure

1. Given: Centralized authentication service returns a user identity for an academic building network access request
2. When: The system receives the identity via integration with centralized authentication

Expected result: Expected: The system validates the identity and applies appropriate role-based policies without requiring manual intervention

Evidence to record: Evidence: External test capture of successful authentication flow and policy application

Test type: positive

Id: TEST-014

Requirement ids: DR-001

Procedure

1. Given: The platform is deployed in an academic building, hostel, laboratory, or administrative office environment
2. When: Network traffic is observed across these physical locations

Expected result: Expected: The system maintains consistent filtering and policy enforcement without degradation in performance or availability

Evidence to record: Evidence: External test capture of network behavior before and after deployment in each location

Test type: analysis

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-002; E-003; E-004

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-004

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-004

14. SDLC recommendation and workflow

Selected model: Agile

Justification: No differentiating input evidence is provided to favor one SDLC model over others. All models receive equal suitability points (40/100) due to absence of factors such as budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification. The decision remains provisional until these unknown factors are clarified.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall implement a secure DNS and web-access filtering mechanism to protect all university users from malicious or unauthorized internet destinations.; FR-002: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events.; FR-003: Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.
2. Review and baseline the source-defined work group: NFR-001: The solution shall be secure, scalable, reliable, and manageable.; DR-001: The platform shall maintain audit logs of policy changes and filtering decisions, integrate with centralized authentication, and support deployment across academic buildings, hostels, laboratories, and administrative offices.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001; DR-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall implement a secure DNS and web-access filtering mechanism to protect all university users from malicious or unauthorized internet destinations.; FR-002: The system shall inspect DNS requests, block access to known malicious domains, apply role-based browsing policies, and provide administrators with a dashboard for reviewing blocked requests and security events.; FR-003: Administrators shall be able to create, update filtering policies, allow approved exceptions, investigate repeated access attempts to blocked resources, and temporarily restrict users when suspicious behavior is detected.
3. Implement the source-defined work group: NFR-001: The solution shall be secure, scalable, reliable, and manageable.; DR-001: The platform shall maintain audit logs of policy changes and

filtering decisions, integrate with centralized authentication, and support deployment acrossacademic buildings, hostels, laboratories, and administrative offices.

4. In an authorized isolated test environment, execute TEST-001: Given: A student attempts to access a known malicious domain When: The DNS resolver receives the request Expected: The system blocks the request.
5. In an authorized isolated test environment, execute TEST-002: Given: A faculty member attempts to access a restricted educational resource When: The system evaluates the request against role-based policy rules Expected: Access is granted according to the faculty policy
6. In an authorized isolated test environment, execute TEST-003: Given: An administrator initiates a dashboard query for blocked requests When: The system retrieves security event logs from its internal store Expected: The dashboard returns a list of recent blocked domains and associated user actions
7. In an authorized isolated test environment, execute TEST-004: Given: An administrator creates a new filtering policy for staff When: The system receives the policy configuration request Expected: The policy is stored and applied to staff user sessions
8. In an authorized isolated test environment, execute TEST-005: Given: The system detects repeated access attempts When: The system detects repeated access attempts Expected: The system triggers an investigation workflow
9. In an authorized isolated test environment, execute TEST-006: Given: An administrator approves an exception for a specific domain When: The system receives the exception request Expected: Access to that domain is permitted for the specified user or group
10. In an authorized isolated test environment, execute TEST-007: Given: A system under operational load with concurrent user sessions When: The system receives a request to scale resource allocation dynamically Expected: The system maintains service availability without degradation in performance or data integrity
11. In an authorized isolated test environment, execute TEST-008: Given: An administrator initiates a configuration change to the security policy When: The system applies the updated policy to active user sessions Expected: The updated security policy is applied to active user sessions
12. In an authorized isolated test environment, execute TEST-009: Given: A component failure is detected in the core service layer When: The system initiates failover to redundant components Expected: The system restores functionality within defined recovery time objectives without data loss
13. In an authorized isolated test environment, execute TEST-010: Given: An operator attempts to modify configuration beyond authorized scope When: The system enforces access control policies during management operations Expected: The system denies the action.
14. In an authorized isolated test environment, execute TEST-011: Given: A system administrator modifies a filtering policy in the university network When: The policy change is applied to the network configuration Expected: An audit log entry is generated recording the timestamp, user identity, action type, and affected resource scope
15. In an authorized isolated test environment, execute TEST-012: Given: A user attempts to access a domain blocked by role-based filtering policies When: The system evaluates the request against active policies and authentication status Expected: The system either blocks the access or allows it based on defined rules, with a decision recorded in audit logs
16. In an authorized isolated test environment, execute TEST-013: Given: Centralized authentication service returns a user identity for an academic building network access request When: The system receives the identity via integration with centralized authentication Expected: The system validates the identity and applies appropriate role-based policies without requiring manual intervention

17. In an authorized isolated test environment, execute TEST-014: Given: The platform is deployed in an academic building, hostel, laboratory, or administrative office environment When: Network traffic is observed across these physical locations Expected: The system maintains consistent filtering and policy enforcement without degradation in performance or availability
18. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
19. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001; DR-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001; DR-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure DNS and web-access filtering system for a university that protects students, faculty, staff, and administrative users from malicious or unauthorized internet destinations.

Id: E-002

Source: description

Quote: The system should inspect DNS requests, block access to known malicious domains, apply different browsing policies based on user roles, and provide administrators with a dashboard for reviewing blocked requests and security events.

Id: E-003

Source: description

Quote: Administrators should be able to create and update filtering policies, allow approved exceptions, investigate repeated attempts to access blocked resources, and temporarily restrict users when suspicious behavior is detected.

Id: E-004

Source: description

Quote: The platform should maintain audit logs of policy changes and filtering decisions, integrate with centralized authentication, and support deployment across academic buildings, hostels, laboratories, and administrative offices.

Id: E-005

Source: description

Quote: The solution should be secure, scalable, reliable, and manageable.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7363

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7357

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.72

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-c1e0239af9dd46bd:2 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 2

Page or section: text chunk 3

Relevance score: 0.7049

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:5 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 5

Page or section: text chunk 6

Relevance score: 0.7107

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: qwen3:4b-instruct-2507-q4_K_M

Data requirements: qwen3:4b-instruct-2507-q4_K_M

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.28

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	TEST-002, TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-003	UC-001, UC-002, TEST-004, TEST-005, TEST-006, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-005	TEST-007, TEST-008, TEST-009, TEST-010, Requirements and verification planning, Agile delivery cycle, Release and handoff
DR-001	E-004	TEST-011, TEST-012, TEST-013, TEST-014, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.28